UNIVERSITY of RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF INFORMATION TECHNOLOGY

END OF SEMESTER I EXAMINATION DURING THE ACADEMIC YEAR 2025-2026

DEPARTMENT: IT **YEAR: II** **SEMESTER: I**

MODULE: IT80361 - Interactive Web Development

DATE: 21/01/2026 **TIME: 2 hours**

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains **FOUR (4)** questions.
2. **Answer THREE (3) Questions only:**
   Question ONE (1) from Section "A" is Compulsory and Answer any TWO (2) from Section "B"
3. No written materials are allowed.
4. No Phones are allowed in the examination room.
5. Do not forget to write your Registration Number.
6. Write all your answers in the booklet provided
7. Do not write any answers on this question paper.

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IT80361 - Interactive Web Development

Duration: 2 hrs. **Maximum Marks: 50**

SECTION A
Compulsory questions: **(Total= 20 marks)**

1.

1.1 Describe how a web browser interacts with a web server during the Request-Response Cycle. Provide examples of HTTP methods used. **[4 marks]**

1.2 Briefly explain the following terms **[4 marks]**

1.2.1 Python Virtual Environment

1.2.2 JWT

1.2.3 QuerySet

1.2.4 REST APIs

1.3 Differentiate the following: **[4 marks]**

1.3.1 PUT method and PATCH method in REST APIs.

1.3.2 Django Project and Django App.

1.4 What would the following command do in a production environment?

```
python manage.py collectstatic
```

[3 marks]

1.5 With the help of a diagram, describe the Django Request-Response Life Cycle, starting from a user entering a URL in the browser to the final HTML response. **[5 marks]**

SECTION B
Answer any two questions: **(2* 15 marks each = 30 marks)**

2. You are developing a Library Management System API using Django REST Framework.

2.1. Define Django models for Book and Author with the following fields: **[5 marks]**

Book: title (CharField), publication_date (DateField), isbn (CharField).

Author: first_name (CharField), last_name (CharField).

Establish a many-to-many relationship between Book and Author.

2.2. Create Django REST Framework serializers for the Book and Author models, ensuring the many-to-many relationship is correctly represented.. **[4 marks]**

2.3. Implement the following API views:

A view to list all books and create a new book entry. **[3 marks]**

A view to retrieve, update, or delete a specific author by their ID. **[3 marks]**

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3.1 Consider a database for an e-commerce platform with Customer and Order tables (where one customer can have many orders)..

3.1.1 Write the Django model code to represent this relationship **[4 marks]**

3.1.2 Write a QuerySet to find all orders placed by a customer with the email 'john.doe@example.com' **[2 marks]**

3.1.3 Write a QuerySet to list all orders that were created in the year 2025 **[2 marks]**

3.2 You are building a blog. The Post model has title, content, author, and created_at fields..

3.2.1 Write a view function post_detail that retrieves a single post by its ID and renders it using a template called detail.html inside a blog directory. **[4 marks]**

3.2.2 In the Django template (detail.html), how would you display the post's title and format the created_at date to show only the month and year?. **[3 marks]**

4.

4.1. In Django, explain how Path Converters (<int:pk>, <slug:slug>) are used in URL routing to capture dynamic data **[5 marks]**

4.2. Use a table to explain the differences between a class based View and a function based view in Django. Provide examples of when you would use each data structure. **[5 marks]**

4.3. Create a class called Product with attributes name, price, and stock_quantity

Implement a method apply_discount(percentage) that reduces the price.

Create a list of three Product objects.

Write a loop to print the names of products that are currently "Out of Stock" (quantity is 0

[5 marks]

ALL THE BEST

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